Venkata Lakshmi Parimala Pasupuleti

Full-Stack Software Engineer · Multifamily Real Estate · Python · TypeScript · React · Data Pipelines · ML Models · PostgreSQL · AWS · GCP · 9yr Production

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Arlington, Virginia, USA (Remote)

PROFESSIONAL SUMMARY

Full-stack software engineer with 9+ years of production experience and 8 years of multifamily real estate domain expertise - the combination Rentana's AI copilot platform needs. I spent 7 years at Entrata building production software for the same NMHC Top 50 property operators Rentana serves: data pipelines normalizing federal and state housing data at scale, complex SQL schemas for portfolio-level financial and eligibility calculations, and backend systems handling billions of records across lease, maintenance, and compliance workflows. That's not adjacent domain knowledge - it's the exact operational context Rentana's revenue intelligence platform is built for. On the AI side: LeaseI is a production AI investigation platform built for regulated real estate workflows, with RAG over regulatory documents and deterministic guardrails for compliance decisions. AgentProbe demonstrates ML-driven evaluation with measurable quality outcomes. I build across the full stack (Python backend, TypeScript/React frontend, PostgreSQL, AWS, GCP), and I work well in small high-ownership environments - at Entrata I was the long-term owner of critical modules, and at Magizhs I took a product from zero to production as sole engineer. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages, Frameworks & AI/ML: Python (9yr), TypeScript, JavaScript, React.js, FastAPI, Flask, LangGraph, LangChain, ML Models (PyTorch, scikit-learn, XGBoost, Random Forest, Logistic Regression), LLM Integration, RAG Pipelines, Agentic Systems, LLM Evaluation, Anomaly Detection, Feature Engineering, Statistical Modeling, Anthropic API, OpenAI API, Prompt Engineering

Data, APIs & Backend: PostgreSQL, MySQL, Oracle, Redis, SQL, NoSQL, SQLite, REST APIs, API Design, GraphQL, Data Pipelines, ETL, Data Normalization, Data Modeling, Async Processing, RabbitMQ, Microservices, Distributed Systems, Performance Optimization, Query Optimization, Indexing, Idempotency, Reliability Engineering

Cloud, DevOps & Testing: AWS (S3, EC2, RDS, Glue, Lambda), GCP (Cloud Storage), Docker, CI/CD (GitHub Actions), Unit Testing, Integration Testing, Regression Testing, Automated Testing, Observability, Structured Logging, Monitoring, Alerting, Design Documents, Technical Mentoring, Code Reviews, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - 0→1 Product Delivery, Small Team

- Took a production AI application from zero to deployed as sole technical owner - designed data pipeline architecture, built RAG retrieval system, evaluation infrastructure, and CI validation gate; worked directly with leadership throughout; delivered in weeks with measurable outcomes driving iteration decisions.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source ML pipelines (PyTorch, scikit-learn, R^2=0.977) on AWS and GCP across 190+ countries; applied Isolation Forest anomaly detection across 3 research pipelines; designed Tableau/ggplot2 dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems.

Entrata - Multifamily Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Multifamily real estate data at production scale** - built and owned data pipelines serving the same NMHC Top 50 property operators Rentana targets: annual federal income-limit data pipelines (few thousand source rows → 180,000 domain records powering rent calculations, eligibility, HAP, and tax-credit decisions for 100K+ housing units); complex PostgreSQL schemas for portfolio-level financial data; ETL processing 1M+ property records; AWS S3/RDS/Glue data infrastructure with 100% compliance on PHI migration.
- **Full-stack product delivery** - shipped 20+ production modules (Python, JavaScript, TypeScript, React.js, PostgreSQL) across 120+ Agile sprints; RabbitMQ async distributed processing with retry logic and idempotency guarantees; 200+ SQL query optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement); GitHub Actions CI/CD with comprehensive unit, integration, and regression test suites.
- **Data quality & reliability** - built structured logging, monitoring, and alerting for data pipeline oversight; zero compliance violations across 7yr ownership; Python/Bash automation eliminating 80% of manual data operations; mentored 5+ engineers; 300+ peer code reviews; internal debugging tools reducing triage time 40%.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management data pipelines (Python, PostgreSQL) for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific document templates.

PROJECTS

Leasel: AI Copilot for Real Estate Compliance - Direct Domain Analog

Real Estate · RAG Pipelines · Agentic · ML · Data Pipelines · Production

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI copilot for real estate compliance operations - a direct working prototype of the AI-native property intelligence platform Rentana is building. **Real estate domain:** built on 8 years of multifamily property management knowledge - lease compliance, income verification, regulatory requirements, portfolio data workflows - the same operational context Rentana's revenue intelligence platform analyzes. **Data pipeline + normalization:** ingestion pipeline normalizing messy third-party lease documents (OCR fallback, section-aware chunking, TF-IDF retrieval over regulatory corpus) - directly analogous to normalizing third-party property data at scale. **ML-driven outcomes:** 93.6% precision, 87.9% F1 across 1,240 test cases; pass@k harness with multi-trial sampling and LLM-as-judge scoring - measurement-driven iteration throughout.

AgentProbe: AI Evaluation & ML Quality Platform

ML Models · LLM Evaluation · Data Pipelines · FastAPI · CI/CD · Open Source

Python · FastAPI · LangGraph · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic API · OpenAI API | github.com/venkatalakshmiparimala/AgentProbe

Production ML evaluation platform demonstrating full-stack data-driven product engineering: 15-endpoint FastAPI backend, React.js + TypeScript frontend, 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency), pass@k multi-trial harness detecting variance in model outputs, CI regression gate (GitHub Actions) blocking quality degradation. ChromaDB data pipeline with embedding-based semantic retrieval. Demonstrates the analytical and ML instrumentation required for an AI copilot platform - metrics-first, data-driven iteration.

CrashClarity: Large-Scale ML Analysis - 104K Records, Predictive Modeling

ML Models · Random Forest · Feature Engineering · Data Pipelines · Python · R

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages | Portfolio

End-to-end ML pipeline on 104K+ real-world records - data ingestion, normalization, feature engineering across weather, road, and temporal variables, Random Forest + Logistic Regression ensemble (85%+ accuracy), and interactive dashboard deployed on GitHub Pages. Demonstrates the "analytical and machine-learning models" capability the Rentana role explicitly calls out: building ML models that distill large-scale messy data into actionable insights - the core of what Rentana's revenue intelligence platform does for portfolio operators.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017